

UNIVERSITY OF MORATUWA

Faculty of Engineering



Registered Module No: MT1070

Thermodynamics Analysis Project

Group A

Batch 23

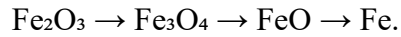
Department of Materials Science and Engineering

Group Members – Group A

Index No	Name with Initials
230265V	Janushan A
230053U	Aranayake A.D.L.
230206T	Gunasekara B.A.T.L.
230433K	Nethsara M.W.V.
230494V	Ponnamperuma K.N.
230710H	Wickramasinghe W.M.C.L.
230368N	Lakshan W.B.T.
230668J	Vishmini I.H.S.
230450J	Outschoorn D.

Summary

This report presents a thermodynamic analysis of reduction reactions occurring in a blast furnace during ironmaking. The study focuses on determining the furnace height zones where major reduction reactions become spontaneous, based on the Gibbs free energy criterion ($\Delta G = \Delta G^\circ + RT \ln Q$). Key reactions considered include the Boudouard reaction and stepwise reduction of iron oxides:



A temperature profile dependent on furnace height and partial pressure models for CO and CO₂ were used to calculate ΔG as a function of height. Phase-specific constants A and B were applied in the expression $\Delta G^\circ = A + B \cdot T$ to account for structural and magnetic transitions in the iron oxides. Graphs of ΔG vs. height were plotted for each reaction.

Results show the temperature and gas composition zones where each reaction becomes thermodynamically favorable. The discussion addresses deviations from real furnace behavior and suggests improvements including the use of real gas profiles, inclusion of hydrogen reduction, and more accurate temperature modeling. The analysis contributes to a deeper understanding of blast furnace efficiency and process control.

Table of Contents

1. Introduction

- 1.1 What is blast furnace?
- 1.2 What we are going to analyze?

2. Fundamentals of Blast Furnace Operation

- 2.1 Overview of Ironmaking Process
- 2.2 Role of Reduction Reactions in the Blast Furnace
- 2.3 Role of fuel Reactions in the Blast Furnace
- 2.4 Importance of Thermodynamic Analysis

3. Thermodynamic Framework

- 3.1 Gibbs Free Energy and Reaction Spontaneity
- 3.2 Effect of Temperature for ΔG
- 3.3 Understanding the Gibbs Free Energy Equation

4. The Boudouard Reaction

- 4.1 Temperature-Dependent Partial Pressures in the Blast Furnace
- 4.2 Temperature Profile in the Furnace

5. Analysis of Key Reduction Reactions

- 5.1 Reduction of Iron Oxides by Carbon Monoxide in a Blast Furnace
- 5.2 Thermodynamic Analysis Procedure
- 5.3 Assumptions and Considerations
- 5.4 Final Thermodynamic Expression

6. Result

7. Discussion

8. Reference

1. Introduction

1.1 What is blast furnace?

The blast furnace (BF) remains the dominant and most efficient industrial reactor for large-scale production of liquid iron, known as pig iron, from iron oxide ores. This process forms the fundamental base of integrated steelmaking worldwide. The blast furnace operates as a tall, continuous, counter-current shaft reactor-typically 30-40 meters in height, with internal volumes ranging from 3000 to 5000 cubic meters in modern installations. Within it, solid raw materials descend while hot reducing gases ascend, enabling thermal and chemical exchange. Iron oxides (primarily Fe_2O_3 and Fe_3O_4) are reduced to metallic iron by carbon monoxide (CO) gas, which is generated in situ by combustion of carbonaceous fuel, typically metallurgical coke. Preheated air (blast), often enriched with oxygen, is injected near the bottom of the furnace through tuyeres, initiating combustion reactions that can generate localized temperatures exceeding 2000°C .

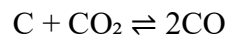
At these high temperatures, the reduced iron melts and separates from non-metallic impurities (gangue and ash), forming two layers at the hearth: molten pig iron, which settles at the bottom due to its higher density, and a floating molten slag layer. The pig iron is periodically tapped and subsequently transferred to basic oxygen furnaces (BOFs) for steel refining. Despite advancements in alternative ironmaking technologies, the blast furnace remains central to steel production due to its scalability, energy efficiency, and process integration. Its operation has been refined over centuries into a highly optimized and continuous system capable of producing over 10,000 tons of hot metal per day with excellent thermal economy.

1.2 What we are going to analysis?

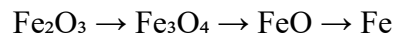
The primary objective of this study is to analyze the height ranges within the blast furnace at which the major reduction reactions are rendered spontaneous. The spontaneity of a chemical reaction is indicated by a negative change in Gibbs free energy ($\Delta G < 0$). Since ΔG is influenced by temperature (T), enthalpy change (ΔH), and entropy change (ΔS), the behavior of reduction reactions is determined by these thermodynamic parameters across the vertical temperature zones of the furnace.

The following reactions are considered in this analysis:

- Boudouard reaction:



- Stepwise reduction of iron oxides:



To assess the spontaneity of these reactions at various furnace heights, the Gibbs free energy change (ΔG) is calculated using the equation:

$$\Delta G = \Delta G^\circ + RT \ln(Q)$$

Where:

- ΔG° is the standard Gibbs free energy change (at 298 K and 1 atm),
- R is the universal gas constant (8.314 J/mol·K),
- T is the absolute temperature in Kelvin,
- Q is the reaction quotient.

By examining how ΔG is varied with temperature, the corresponding height zones of spontaneity for each reaction are identified. In this way, thermodynamic favorability is linked to specific furnace regions. This information is considered valuable for enhancing the understanding of furnace operation, and it is expected to contribute to improved efficiency and reduced fuel consumption.

2. Fundamentals of Blast Furnace Operation

2.1 Overview of Ironmaking Process

The blast furnace (BF) is the dominant industrial technology to produce liquid iron (pig iron) from iron oxide ores, the key starting point in integrated steelmaking. It operates as a continuous, counter-current reactor: solid burden material descends, while hot reducing gases ascend.

Key Inputs:

- **Iron-bearing Materials:** Agglomerated iron ore is used to ensure good permeability and reducibility.
- **Fuel & Reductant:** Metallurgical coke is employed as the principal source of carbon, energy, and structural support.
- **Fluxes:** Limestone and/or dolomite are added to react with gangue minerals and ash, forming slag.
- **Blast:** Preheated air (1000-1300°C), usually enriched with oxygen, is supplied via tuyeres.
- **Auxiliary Fuels (Optional):** Pulverized coal, natural gas, or oil may be injected through the tuyeres to supplement coke.

Process Overview:

Coke is ignited by the hot blast at the tuyeres, resulting in the generation of reducing gases (CO, H₂) and temperatures exceeding 2000°C. These gases are made to rise, heating and reacting with the descending burden as it is processed through various temperature zones:

- **Upper Shaft ($\leq 400^{\circ}\text{C}$):** The burden is dried and preheated.
- **Mid-Shaft (400-900°C):** Iron oxides are indirectly reduced ($\text{Fe}_2\text{O}_3 \rightarrow \text{Fe}_3\text{O}_4 \rightarrow \text{FeO} \rightarrow \text{Fe}$) by CO and H₂, while limestone is calcined.
- **Lower Shaft/Bosh (900-1500°C):** Direct reduction is carried out ($\text{FeO} + \text{C} \rightarrow \text{Fe} + \text{CO}$), and the burden is melted, with slag being formed.
- **Hearth/Tuyere Zone ($> 1500\text{-}2000^{\circ}\text{C}$):** Full coke combustion is achieved, and the separation of slag and molten metal is completed.

Outputs:

- Molten Pig Iron (93-95% Fe, 4-4.5% C, Si, Mn, P, S) is tapped and is sent to the Basic Oxygen Furnace (BOF) for steelmaking.
- Molten Slag ($\text{CaO-SiO}_2\text{-Al}_2\text{O}_3\text{-MgO}$) is tapped and is utilized, often in cement production.
- Top Gas ($\text{CO, CO}_2, \text{N}_2, \text{H}_2$) is cleaned and is used in blast preheating and power generation.

This process is recognized for its massive production scale (>10,000 tonnes of hot metal per day), high thermal efficiency, and the ability to operate continuously for 15-20 years.

2.2 Role of Reduction Reactions in the Blast Furnace

Reduction reactions are the basic chemical conversions by which iron oxides (Fe_2O_3 , Fe_3O_4 , FeO) are reduced to metallic iron within the blast furnace. The reduction is carried out almost entirely by gaseous reduction (indirect reduction) in the top and middle shaft portions ($400\text{--}900^\circ\text{C}$), where carbon monoxide (CO) and hydrogen (H_2), generated in the lower parts of the furnace, are used as reducing agents. Sequential reduction $\text{Fe}_2\text{O}_3 \rightarrow \text{Fe}_3\text{O}_4 \rightarrow \text{FeO} \rightarrow \text{Fe}$ is required, with the reduction of FeO being considered the most energy-intensive step.

Indirect reduction is regarded as thermally efficient, as the chemical energy of the ascending gas is utilized, but it is made dependent on gas-solid contact and the reducibility of the ore. Below approximately 900°C , in the bosh and lower shaft, direct reduction by solid-state reactions is initiated, mainly for FeO reduction ($\text{FeO} + \text{C} \rightarrow \text{Fe} + \text{CO}$). Though direct reduction is characterized by high coke (carbon) consumption and an endothermic nature, it is considered essential for completing the reduction process when gaseous reductants are rendered thermodynamically inefficient. Additionally, significant amounts of CO gas are generated during this stage, which is used for upward reduction.

The proportion of direct to indirect reduction is known to have a substantial influence on overall furnace efficiency, coke consumption rate, and productivity. The Boudouard reaction ($\text{C} + \text{CO}_2 \rightleftharpoons 2\text{CO}$), which is conducted in the high-temperature reserve zone ($\sim 900\text{--}1000^\circ\text{C}$), is identified as a key process in generating the primary gaseous reductant (CO), which is consumed during indirect reduction above.

Reduction of direct reduction requirements (with higher coke consumption per unit of iron) is pursued as an optimization goal, typically achieved by maximizing indirect reduction through increased ore reducibility, improved gas utilization, and optimized burden distribution. Effective reduction is expected to prevent excess residual oxygen in the descending burden before melting, allowing optimal coke consumption, regulating the furnace's thermal profile, and facilitating continuous production of high-quality, low-oxygen molten iron, which is tapped into the hearth.

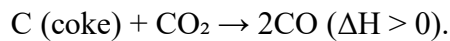
Hence, the kinetics, degree, and location of these reduction reactions are regarded as critical factors in determining the cost and operational stability of the blast furnace.

2.3 Role of fuel Reactions in the Blast Furnace

Fuel reactions are regarded as the sources of reductant and heat required for the blast furnace to operate. The principal reaction is carried out at the tuyeres, where metallurgical coke is combusted with pre-heated (often oxygen-enriched) air in an exothermic reaction:
$$\text{C} + \text{O}_2 \rightarrow \text{CO}_2 + \text{Heat} (\Delta H < 0).$$

Temperatures exceeding 2000 °C are thereby produced, by which hearth materials are melted and a high-temperature zone for slag–metal separation is established. The CO₂ that is generated is subsequently made to react with coke within the raceway and bosh through the endothermic

Boudouard reaction:



Although a large amount of heat is absorbed, the chief gaseous reducing agent, CO, is produced and is allowed to ascend to reduce iron oxides in the shaft. Coke is thus assigned three critical functions: it is employed as an energy source, it is relied upon to maintain burden permeability, and it is used as a chemical precursor for reduction gas.

Additional fuels-natural gas, pulverized coal, or oil-are injected through the tuyeres to supplement coke. Upon combustion (e.g., $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{Heat}$) and pyrolysis (e.g., $\text{CH}_4 \rightarrow \text{C} + 2\text{H}_2$), heat and reducing agents (CO, H₂) are introduced. Hydrogen is believed to enhance reduction kinetics in low-temperature zones, but the gas volume is increased, so blast parameters must be carefully controlled to secure stable flame temperature and gas flow.

The combined effect of all fuel combustion is understood to dictate the chemical and thermal reserve zones of the furnace: exothermic combustion at the base is relied upon to provide the energy reservoir, and a continuous supply of CO from the Boudouard reaction is ensured for indirect reduction higher in the shaft. Proper equilibrium of these phenomena is considered crucial for controlling coke consumption, furnace productivity, and overall efficiency.

2.4 Importance of Thermodynamic Analysis

Thermodynamic analysis is considered indispensable for determining the feasibility and spontaneity ($\Delta G < 0$) of key reactions (e.g., iron oxide reduction, Boudouard reaction) across the furnace's temperature zones. Critical equilibrium thresholds, such as the Boudouard equilibrium temperature (~ 950 - 1000°C), are identified, which is known to define the "thermal reserve zone" where CO_2 is efficiently converted to CO -thus optimizing the generation of reductant gas for indirect reduction.

Through this analysis, the theoretical minimum energy requirement (e.g.:coke rate) is established, and the dominance of specific reactions in certain zones (e.g., indirect reduction in the shaft vs. direct reduction in the bosh) is explained.

In practice, thermodynamics is used to guide operational optimization in several ways:

- Slag composition is designed (using phase diagrams) for effective impurity removal and appropriate viscosity.
- Raw materials are selected based on ore reducibility and flux basicity, and blast parameters (such as O_2 enrichment and humidity) are adjusted accordingly.
- Issues such as scaffold formation or abnormal fuel consumption are diagnosed and addressed.

By providing the scientific foundation for energy efficiency, coke minimization, and process stability, thermodynamic analysis is regarded as essential for both furnace design and daily operation.

3. Thermodynamic Framework

3.1 Gibbs Free Energy and Reaction Spontaneity

The Gibbs free energy (G) is one of a thermodynamic potential that predicts the spontaneity of a process at constant temperature (T) and pressure (P). It is defined as:

$$G = H - TS$$

Where,

- H is enthalpy
- T is absolute temperature
- S is entropy.

○ Gibbs Free Energy Change (ΔG)

For a reaction:

$$\Delta G = \Delta H - T\Delta S$$

$$dG = dH - TdS - SdT$$

$$dG = dH - TdS \text{ (If } T \text{ constant)}$$

$$dG = \sum \mu_i dn_i$$

For the reaction of Gibbs:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

ΔG = Gibbs free energy change at non-standard conditions (J/mol or kJ/mol)

ΔG° = Standard Gibbs free energy change (at 298 K and 1 atm) (J/mol or kJ/mol)

R = Universal gas constant = 8.314 J/mol·K

T = Absolute temperature in Kelvin (K)

Q = Reaction quotient

Assumptions

- Constant Temperature (T)
- Constant Pressure (P)
- Only reversible (idealized) processes considered
- No work other than PV work
- ΔH : Enthalpy change (heat absorbed/released).
- ΔS : Entropy change (disorder).

Spontaneity Criterion,

- $\Delta G < 0$: Spontaneous
- $\Delta G > 0$: Non-spontaneous
- $\Delta G = 0$: Equilibrium

3.2 Effect of Temperature for ΔG

The temperature dependence is captured by the Gibbs-Helmholtz Equation.

- The $-T\Delta S$ term means ΔG changes linearly with temperature if ΔH and ΔS are constant.
- The sign and magnitude of ΔS determine the direction of the change:
 - Positive ΔS ($\Delta S > 0$): Favors spontaneity. As T increases, $-T\Delta S$ becomes more negative, making ΔG more negative (more spontaneous).
 - Negative ΔS ($\Delta S < 0$): Opposes spontaneity. As T increases, $-T\Delta S$ becomes more positive, making ΔG less negative or more positive (less spontaneous).
- Temperature can change the sign of ΔG . Reactions with $\Delta H > 0$ (endothermic) and $\Delta S > 0$ become spontaneous only above a certain temperature.
- Reactions with $\Delta H < 0$ (exothermic) and $\Delta S < 0$ become spontaneous only below a certain temperature.

3.3 Understanding the Gibbs Free Energy Equation

$$\Delta G^\circ = \begin{cases} A_1 + B_1T ; T_0 < T < T_1 \\ A_2 + B_2T ; T_1 < T \end{cases}$$

Where:

- T = Temperature (in Kelvin), which is the independent variable in the equation
- ΔG° : Standard Gibbs free energy change (kJ/mol)
- A_1, A_2 : Intercept terms (linked to ΔH° , enthalpy change).
- B_1, B_2 : Slope terms (linked to $-\Delta S^\circ$, entropy change).

- T_1 : Phase transition temperature
- T_0 : 298K

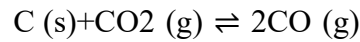
The stepwise formulation $\Delta G^\circ = A + BT$ is necessary because phase transitions introduce discontinuities in thermodynamic properties. Consequently, the coefficients A (reflecting enthalpy changes) and B (representing entropy effects) assume distinct values across different phase domains, requiring this segmented functional representation.

4. The Boudouard Reaction

The Boudouard Reaction

Section: Thermodynamic Analysis of the Boudouard Reaction in Ironmaking

The Boudouard reaction, represented by the equilibrium:



plays a pivotal role in high-temperature metallurgical processes, especially within iron blast furnaces. This reaction determines the concentration of carbon monoxide (CO), the primary reducing agent for iron oxides.

The Gibbs free energy change is given by:

$$\Delta G = \Delta H - T\Delta S$$

the effect of temperature-dependent heat capacities should be incorporated. The standard Gibbs energy equation assumes constant enthalpy and entropy, but in reality, and vary with temperature. These variations can be quantified by:

$$\Delta H_T = \Delta H^\circ + \int_{298}^T \Delta C_p \cdot dT$$

$$\Delta S_T = \Delta S^\circ + \int_{298}^T \frac{\Delta C_p}{T} \cdot dT$$

where .

$$\Delta C_p = 2C_p^{CO} - C_p^{CO_2} - C_p^C$$

Representative values such as , $C_p^{CO} = 29.1 + 0.0011T$, $C_p^{CO_2} = 37.1 + 0.0015T$ and $C_p^C = 8.5$, can be used for practical estimation.

$$\Delta C_p = 12.6 + 0.0007T$$

$$\Delta G = \Delta H^\circ + \int_{298}^T \Delta C_p \cdot dT - T(\Delta S^\circ + \int_{298}^T \frac{\Delta C_p}{T} \cdot dT)$$

$$\Delta G = \Delta H^\circ + \int_{298}^T 12.6 + 0.0007T \cdot dT - T(\Delta S^\circ + \int_{298}^T \frac{12.6 + 0.0007T}{T} \cdot dT)$$

$$\Delta G = \Delta H^\circ + \int_{298}^T \Delta C_p \cdot dT - T(\Delta S^\circ + \int_{298}^T \frac{\Delta C_p}{T} \cdot dT)$$

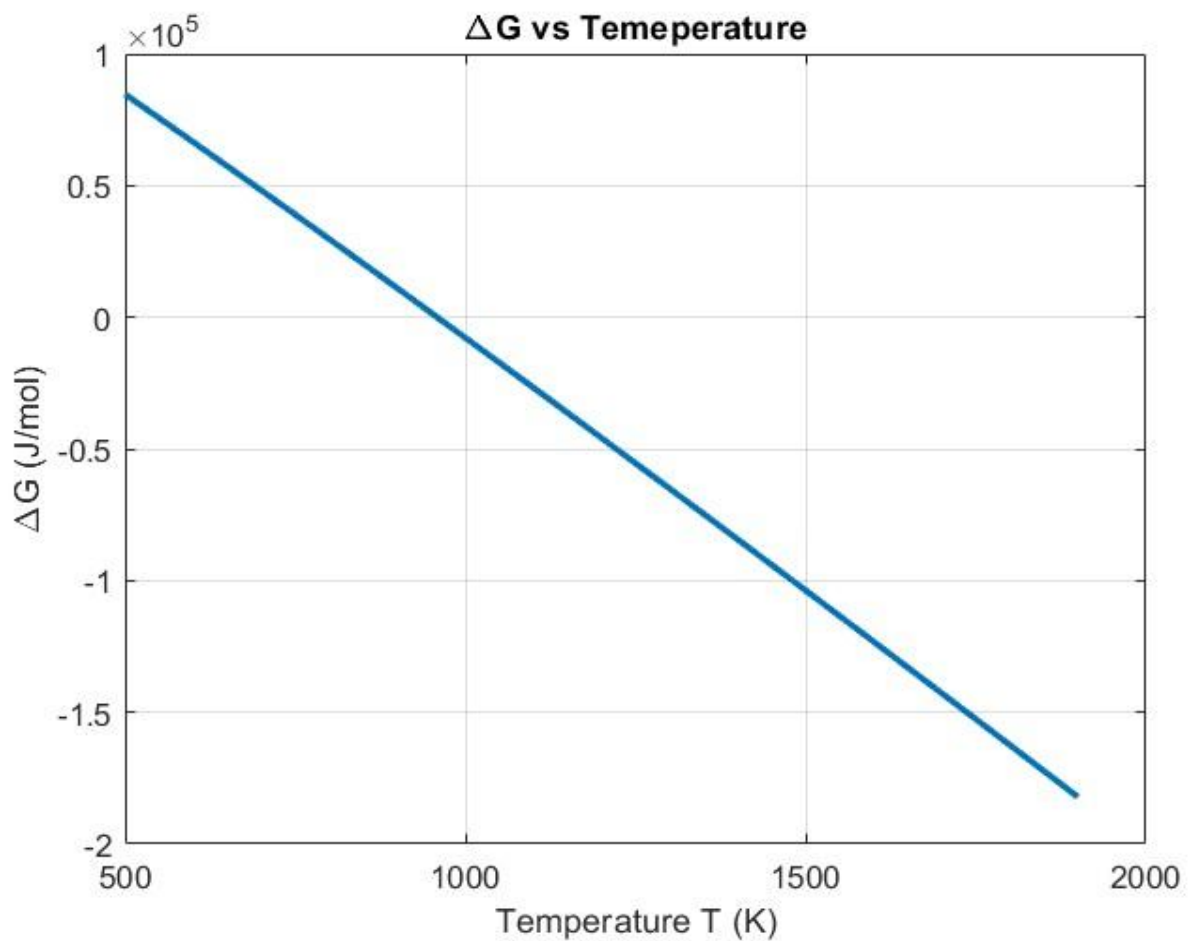
$$\Delta G(T) = A + BT + C T^2 + D \cdot T \ln(T)$$

$$A = -3785.67 + \Delta H^\circ$$

$$B = 84.38 - \Delta S^\circ$$

$$C = 3.5 \times 10^{-4}$$

$$D = -12.6$$



4.1 Temperature-Dependent Partial Pressures in the Blast Furnace

Within the blast furnace, the partial pressures of CO and CO₂ vary significantly with temperature due to gas-phase reactions and equilibrium. Empirical relationships model these partial pressures as temperature-dependent functions:

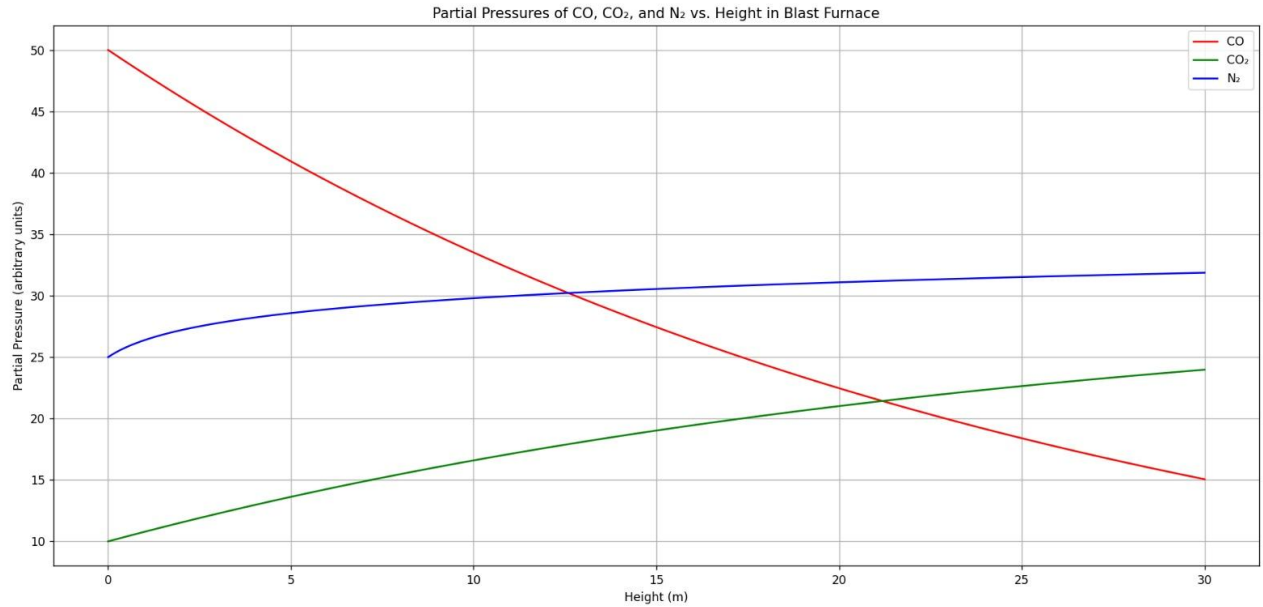


Figure 1 [5],[6]

The function was derived through curve fitting of the above graph.

$$P_{CO_2} = A_1 \cdot (1 - e^{B_1 h}) + C_1$$

$$P_{CO} = A_2 \cdot e^{B_2 h} + C_2$$

$$A_1 = 23$$

$$A_2 = 63$$

$$B_1 = -0.032$$

$$B_2 = -0.028$$

$$C_1 = 9$$

$$C_2 = -12$$

4.2 Temperature Profile in the Furnace

The vertical temperature variation along the furnace height h can be modelled using the following empirical expression:

$$T(h) = T_{Top} + (T_{Base} - T_{Top}) \left(1 - \frac{h}{y}\right)^{1.5}$$

Where:

- $T(h)$: Temperature at height h in Kelvin
- T_{Top} : Temperature at top of the furnace
- T_{Base} : Temperature at base of the furnace
- h : Height from the tuyere level (in meters)
- y : Total height of the furnace (in meters)

This equation shows a non-linear increase in temperature as one moves downward in the furnace, reaching its maximum at the bottom (near the tuyeres) due to combustion of coke with the hot air blast. The curve follows a power law, reflecting real furnace conditions better than linear approximations.

$$T(h) = \left[500 + (1400) \left(1 - \frac{h}{30}\right)^{1.5} \right]$$

$$T_{Top} = 400\text{K}$$

$$T_{Base} = 1900\text{K}$$

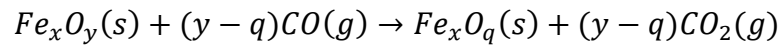
$$y = 30\text{m}$$

These values are used to analyze the blast furnace.

5. Analyze of Key Reduction Reactions

5.1 Reduction of Iron Oxides by Carbon Monoxide in a Blast Furnace

The reduction of iron oxides (Fe_xO_y) by carbon monoxide (CO) in a blast furnace is a fundamental step in iron production. This reduction typically proceeds through several intermediate stages, which can be generally represented by the reaction:



The thermodynamic favorability of this reaction at temperature T (in Kelvin) is evaluated using the Gibbs free energy change (ΔG):

$$\Delta G = \Delta G^0 + RT \ln Q(h)$$

Where:

- ΔG^0 is the standard Gibbs free energy change at temperature T ,
- $R=8.314 \text{ J/mol.K}$ is the universal gas constant,
- $Q(h)$ is the reaction quotient, defined by the partial pressures of gaseous species:

$$Q(h) = \left(\frac{P_{CO_2}}{P_{CO}} \right)^{y-q}$$

$$Q(h) = \left(\frac{63e^{-0.028h} - 12}{32 - 23e^{-0.032h}} \right)^{y-q}$$

Since iron oxides and iron monoxide are solids, their activities are taken as unity and do not appear in the expression for Q .

A negative value of ΔG indicates the reaction is thermodynamically spontaneous under those conditions.

As increases, the behavior of is mainly influenced by the exponential terms in both numerator and denominator. Typically, the CO_2 pressure approaches a constant value, while the CO pressure keeps increasing with height. As a result, the overall ratio decreases. This suggests that at greater heights,

the concentration of CO becomes significantly higher than that of CO₂, indicating a shift in gas equilibrium or reaction conditions that favors the formation or presence of CO over CO₂.

5.2 Thermodynamic Analysis Procedure

To analyze the reaction spontaneity over a temperature range (500 K to 1900 K):

1. Obtain A, B values for the reaction at each temperature from thermodynamic data sources.
2. Calculate P_{CO} and P_{CO_2} at each temperature using the empirical models.
3. Compute the reaction quotient Q and then the Gibbs free energy change ΔG .
4. Plot ΔG versus temperature to identify the temperature ranges where the reduction is spontaneous ($\Delta G < 0$).
5. Plot ΔG versus Height to identify the height ranges where the reduction is spontaneous ($\Delta G < 0$)

5.3 Assumptions and Considerations

- Gaseous species behave as ideal gases.
- The system operates at constant pressure, typical of blast furnace conditions. Activities of solid phases (iron oxides and metallic iron) are considered to be unity.
- Empirical constants must be accurately determined from furnace-specific data for precise modeling.
- Real furnace gases contain other components (e.g., nitrogen, hydrogen), which can influence gas partial pressures and reaction equilibria but are not accounted for in this simplified model.

5.4 Final Thermodynamic Expression

Given the temperature distribution in the furnace modeled by:

$$\Delta G = \Delta G^0 + RT \ln Q(h)$$

The complete expression for Gibbs free energy becomes:

$$T(h) = \left[500 + (1400) \left(1 - \frac{h}{30} \right)^{1.5} \right]$$

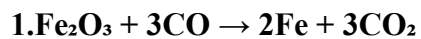
$$Q(h) = \left(\frac{63e^{-0.028h} - 12}{32 - 23e^{-0.032h}} \right)^{y-q}$$

By using these functions we can get generalized function for Gibbs free energy as function of h

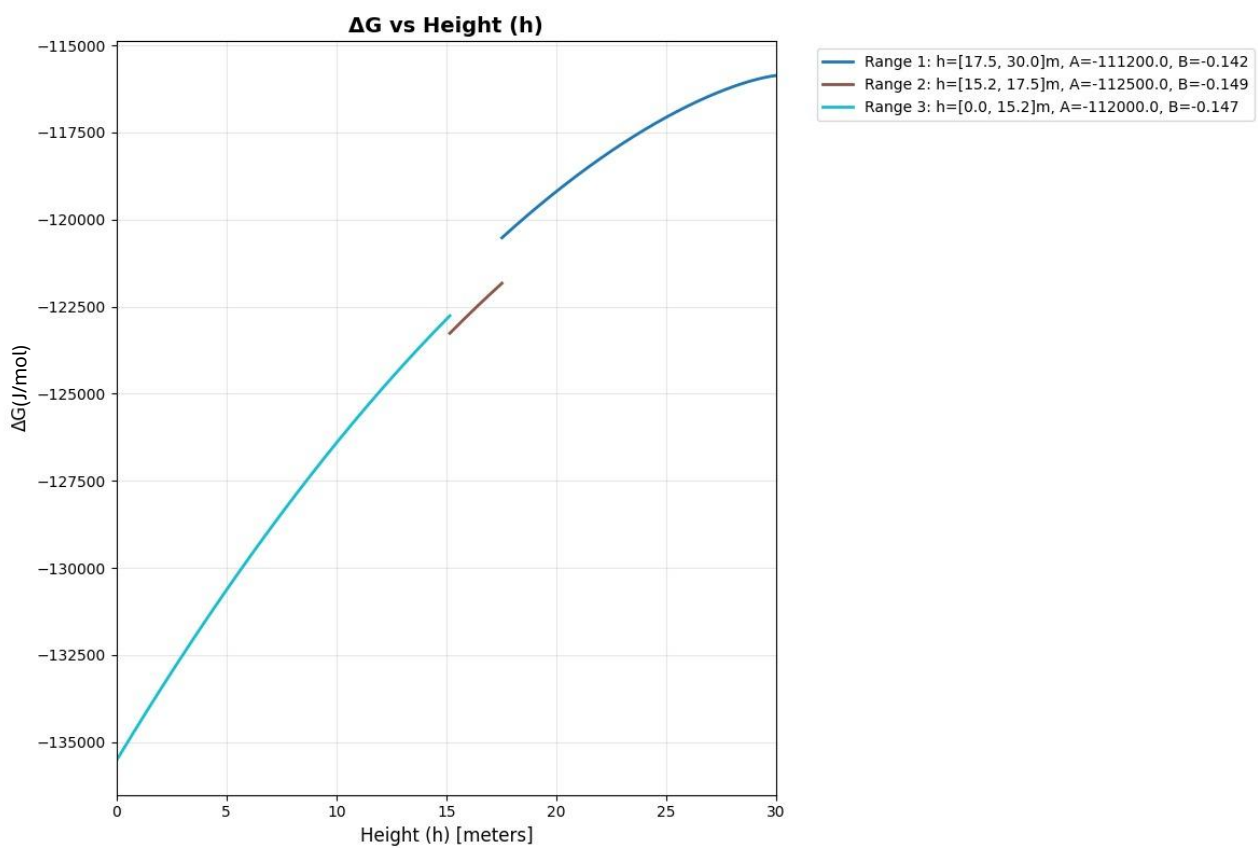
$$\Delta G = A + B.T(h) + R.T(h).ln[Q(h)]$$

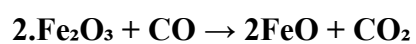
A , B = CONSTANTS

6. Results

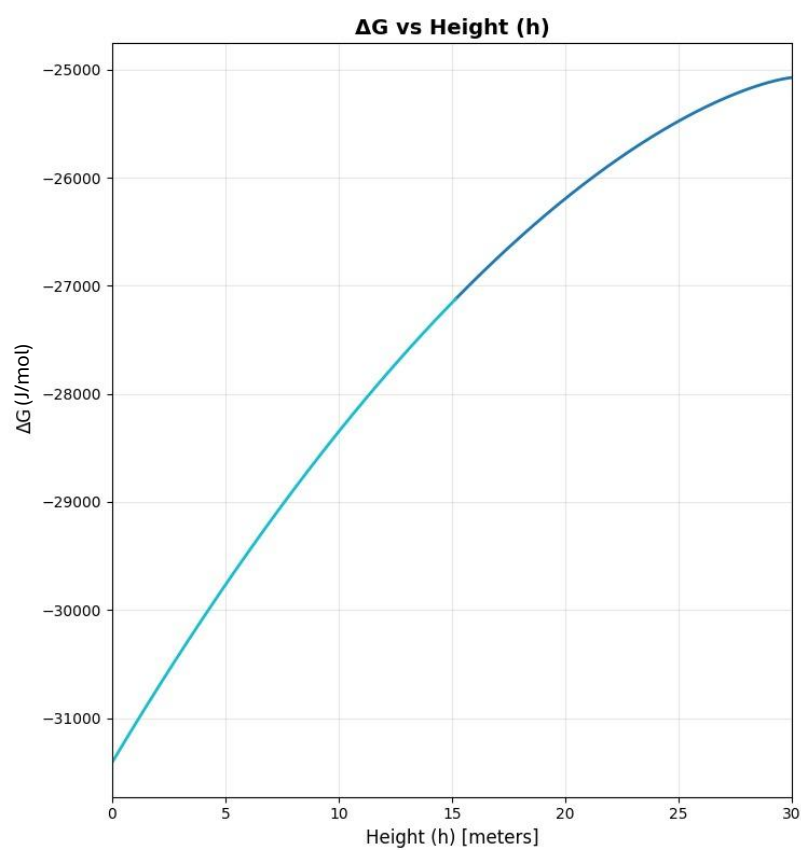


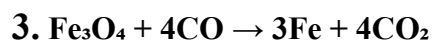
Temp Range (K)	Phase Characteristics	A (J/mol)	B (J/mol·K)
500-848	Antiferromagnetic phase	-111,200	-0.142
848-960	Paramagnetic phase	-112,500	-0.149
960-1800	Stable hematite	-112,000	-0.147



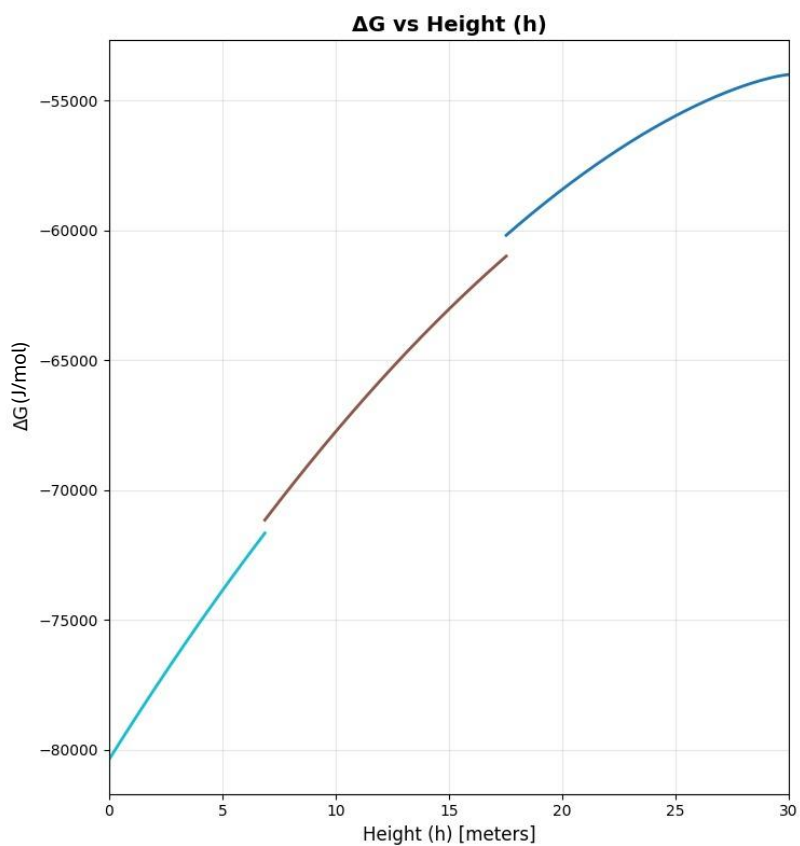


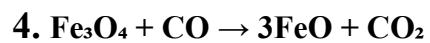
Temp Range (K)	Phase Characteristics	A (J/mol)	B (J/mol·K)
500-960	Magnetic transitions	-23,500	-0.085
960-1800	Stable reduction	-23,500	-0.085



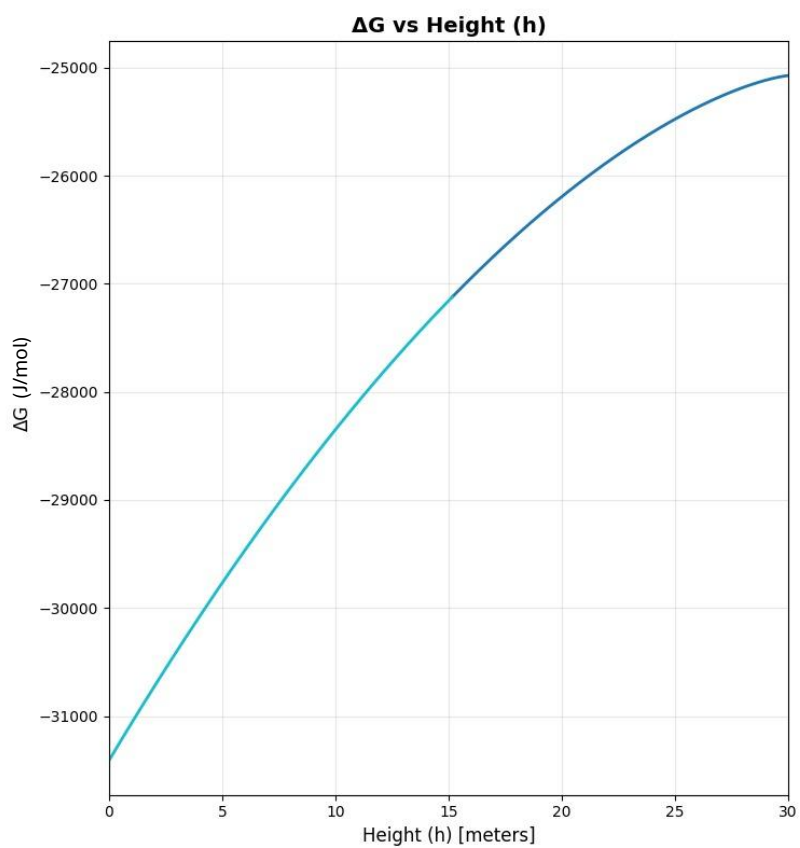


Temp Range (K)	Phase Characteristics	A (J/mol)	B (J/mol·K)
500-848	Monoclinic structure	-47,800	-0.132
848-1420	Cubic structure	-48,600	-0.138
1420-1870	Paramagnetic phase	-49,100	-0.141



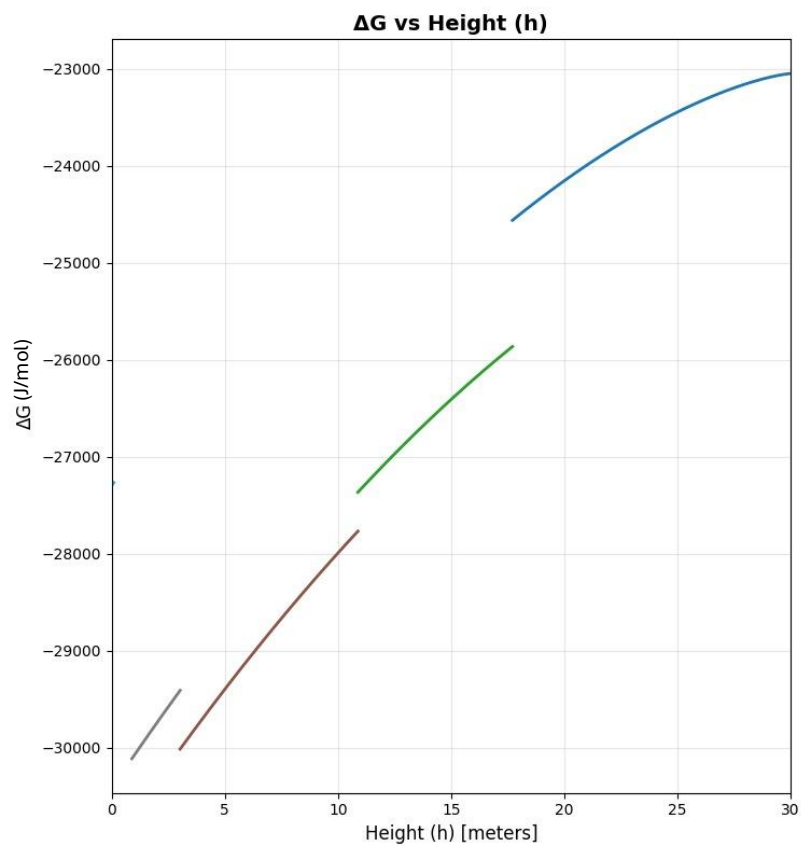


Temp Range (K)	Phase Characteristics	A (J/mol)	B (J/mol·K)
570-840	Metastable wüstite	-15,200	-0.062
840-1800	Stable wüstite	-15,200	-0.062



5. $\text{FeO} + \text{CO} \rightarrow \text{Fe} + \text{CO}_2$

Temp Range (K)	Iron Phase	A (J/mol)	B (J/mol·K)
570-840	α -Fe (BCC) formation	-21,500	-0.035
840-1185	α -Fe (BCC) stable	-22,800	-0.038
1185-1667	γ -Fe (FCC) formation	-23,200	-0.040
1667-1811	δ -Fe (BCC) formation	-22,600	-0.037
>1811	Liquid iron	-19,500	-0.025



Phase Change Reference Temperatures

Material	Transition Temperature (K)	Phase Change Description
Fe ₂ O ₃	848	Magnetic: AFM → Paramagnetic
Fe ₂ O ₃	960	Minor structural adjustment
Fe ₃ O ₄	848	Verwey transition: Monoclinic → Cubic
Fe ₃ O ₄	1420	Magnetic: Ferrimagnetic → Paramagnetic
FeO	570	Thermodynamic stability threshold
FeO	840	Cubic structure stabilization
Fe	1043	α-Fe: Ferromagnetic → Paramagnetic
Fe	1185	α-Fe (BCC) → γ-Fe (FCC)
Fe	1667	γ-Fe (FCC) → δ-Fe (BCC)
Fe	1811	Melting: δ-Fe → Liquid

Implementation Notes:

1. Use height-dependent temperature profile for each reaction zone
2. For $h < 30\text{m}$, use tuyere-level values ($h = 30\text{m}$)
3. Gas composition terms optimized for standard blast furnace conditions
4. For liquid iron ($>1811\text{K}$), carburization dominates reaction kinetics

7. Discussion

- To improve the temperature model of a blast furnace, it is essential to move beyond simplified assumptions and incorporate measured temperature profiles or adopt more complex zone-based models that reflect the actual thermal gradients within the furnace. Real blast furnaces exhibit peak temperatures in the range of approximately 1650-2000 °C, particularly near the raceway, and these values should be used to validate the model against credible sources such as operational data or references like Britannica.com. If an empirical approach is preferred, relying on a fixed 1.5 exponent curve may oversimplify the system; instead, fitting the model to real blast furnace temperature data provides a more accurate and representative thermal profile. This refinement allows for more precise predictions of reaction rates and zone behaviors, ultimately enhancing the fidelity of the simulation.
- To accurately model the thermodynamics of a blast furnace with multiple reducing agents, it is important to account for the presence of hydrogen from sources such as injected fuels (e.g., oil or natural gas). In such cases, the reduction reactions are not limited to the CO/CO₂ system; the H₂/H₂O equilibrium must also be included in the Gibbs free energy (ΔG) calculations. Ignoring the H₂/H₂O system can lead to significant errors in predicting the reduction potential, especially in regions where hydrogen becomes a dominant reducer. By incorporating both CO/CO₂ and H₂/H₂O equilibria, the model can more accurately reflect the competition between reducing agents and their respective efficiencies at different furnace zones and temperatures, resulting in a more realistic representation of reduction behavior and furnace performance
- To enhance the reliability of the blast furnace model, it is important to recognize that the partial pressures of CO and CO₂ vary significantly across different zones within the furnace. These spatial variations arise due to non-uniform gas flow, temperature gradients, and local reaction kinetics. As a result, using a single set of partial pressure values throughout the furnace can lead to inaccurate predictions. In practice, the partial pressures measured in one zone may not match those in another, and thus the exact values cannot be uniformly applied. This reinforces the need for

either direct calibration using real probe or plant data or for calculating gas compositions through equilibrium equations tailored to each furnace height. Incorporating zone-specific gas-phase compositions aligned with the degree of solid reduction ensures the model captures realistic process behavior and adheres to both thermodynamic principles and observed operating data.

8. References

- [1] J. G. Peacey and W. G. Davenport, *The Iron Blast Furnace: Theory and Practice*. Oxford, U.K.: Pergamon Press, 1979.
- [2] E. T. Turkdogan, *Fundamentals of Steelmaking*. London, U.K.: Institute of Materials, 1996.
- [3] I. Barin, *Thermochemical Data of Pure Substances*, 3rd ed. Weinheim, Germany: Wiley-VCH, 1993.
- [4] L. S. Darken, "Thermodynamics of Steelmaking," in *Thermodynamics in Metallurgy*, Columbus, OH, USA: American Society for Metals, 1967, pp. 1–45.
- [5] M. Sasabe, K. Kobayashi, M. Yoshikawa, and K. Suzuki, "Direct measurement of oxygen partial pressure in the experimental blast furnace," *ISIJ Int.*, vol. 22, no. 10, pp. 794-798, 1982, doi: 10.2355/isijinternational1966.22.10_794
- [6] C. Bodsworth and H. B. Bell, *The Physical Chemistry of Iron and Steel Manufacture*, 2nd ed. London, U.K.: Longman, 1972. [Online]. Available: <https://archive.org/details/physicalchemistr0000coli>
- [7] Y. Omori, "Blast furnace phenomena and modelling," *ISIJ Int.*, vol. 27, no. 6, pp. 425-438, 1987, doi: 10.2355/isijinternational1966.27.425.
- [8] P. R. Austin, H. Nogami, and J. Yagi, "A mathematical model of four phase motion and heat transfer in the blast furnace," *ISIJ Int.*, vol. 37, no. 8, pp. 748-755, 1997, doi: 10.2355/isijinternational.37.748.
- [9] D. Fu, Y. Chen, and M. Chen, "Recent progress on ironmaking reaction mechanisms in blast furnace," *Metall. Mater. Trans. B*, vol. 51, no. 4, pp. 1479-1492, 2020, doi: 10.1007/s11663-020-01853-5.
- [10] E. T. Turkdogan, *Fundamentals of Steelmaking*, 1st ed. London, U.K.: Institute of Materials, 1996.
- [11] R. J. Fruehan (Ed.), *The Making, Shaping and Treating of Steel: Ironmaking Volume*, 11th ed. Pittsburgh, PA, USA: AISE Steel Foundation, 1998.
- [12] H. Saxén and F. Pettersson, "Mathematical modeling of the blast furnace process: State of the art and recent developments," in *Proc. 4th Int. Cong. Sci. Technol. Ironmaking*, Osaka, Japan, 2006, pp. 22-26.
- [13] M. Geerdes, R. Chaigneau, and I. Kurunov, "Modern blast furnace ironmaking: An introduction," in *Proc. AISTech Conf.*, Indianapolis, IN, USA, 2015, pp. 1-10.
- [14] International Iron Metallurgy Association, "Blast furnace ironmaking," Tech. Rep. IIMA-001, 2019. [Online]. Available: <https://www.metallics.org>

[15] J. A. de Castro, "Three-dimensional mathematical modeling of the blast furnace," Ph.D. dissertation, RWTH Aachen Univ., Aachen, Germany, 2003.